



The Graph Theory Connection in Competitive Mathematics

CPMSoc Mathematics

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Attendance

Please scan the QR code; and let us know about any dietary requirements.



(For the pizza!)



Outline

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Why Do We Care About Graph Theory

1. Models real life problems such as friending, road networks and matching.

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2. Applied in optimisation problems (shortest path, maximum flow).
3. Tool to simplify / visualise Competitive Mathematics and Competitive Programming problems.

Revision

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- Collection of vertices v , connected by edges e where $v \in V$, $e \in E$

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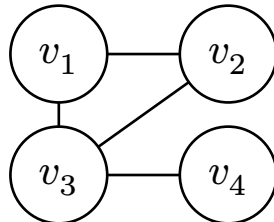
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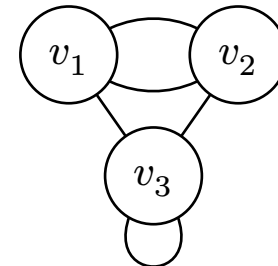
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Multigraph



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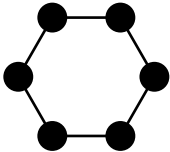
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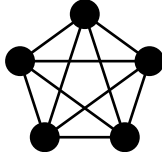
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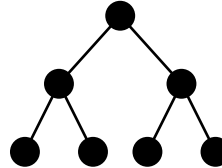
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 C_6 

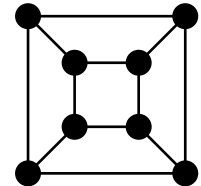
$$|V| = n, |E| = n$$

 K_5 

$$|V| = n, |E| = \frac{n(n-1)}{2}$$

 T_7 

$$|V| = n, |E| = n - 1$$

 Q_3 

$$|V| = 2^n, |E| = n \cdot 2^{n-1}$$

Graph Types

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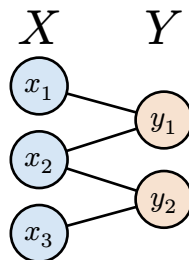
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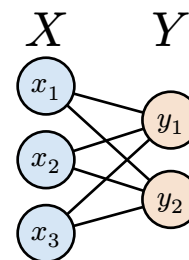
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Bipartite



Complete $K_{3,2}$



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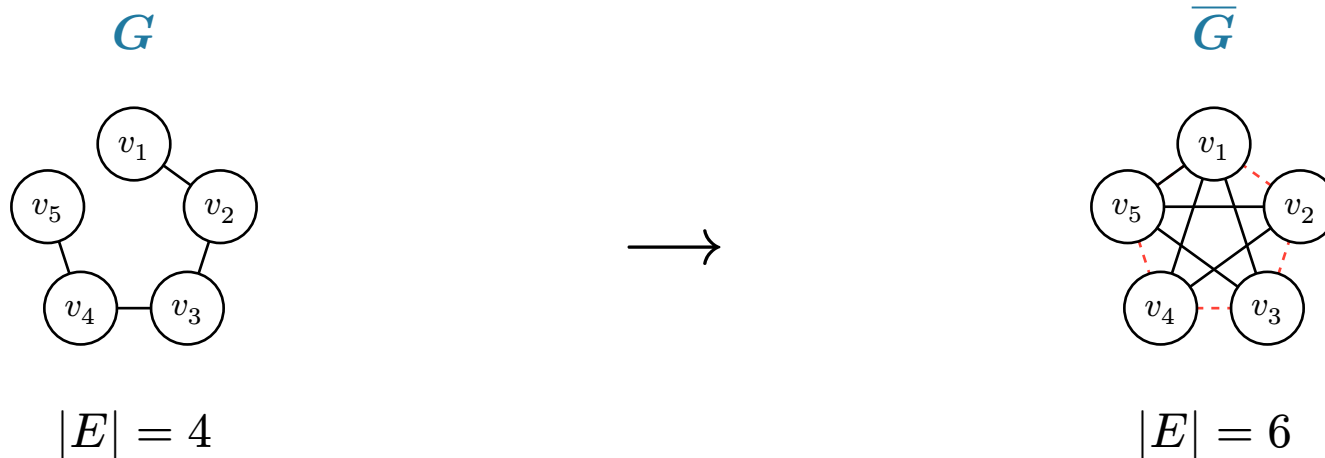
- **Base Case:** Colouring C_3 with two colours always results in two adjacent vertices having the same colour.
- **Inductive Step:** Assume it is not true for $n = 2k + 1$. For contradiction, suppose it is possible for $n = 2k + 3$. Removing two adjacent vertices of different colours gives a colouring of $2k + 1$, a contradiction.

Graph Notation

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$$|E(G)| + |E(\overline{G})| = |E(K_n)| = 10$$

Graph Notation

- **Subgraph:** A subgraph G' is a graph with $V(G') \subseteq V(G)$ and same for the edges. Note that G' must still be a graph, so all endpoints in $E(G')$ must be in $V(G')$, but we can have extra vertices.

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Pick some vertices $S = V(G')$, keep only the edges from $E(G)$ with both endpoints in S and delete everything else. We notate this induced subgraph to be $G[S]$.

Handshaking Lemma

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- Simplifying, we have $|E| = n \cdot 2^{n-1}$.

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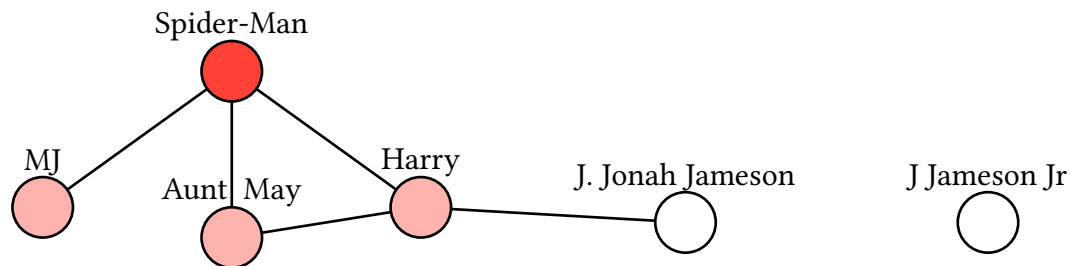
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*The transformation of a graph X into a graph Y where a specified property stays unchanged (**an invariant**)*

$$f : X \rightarrow Y \text{ such that } p(X) = p(f(X)),$$

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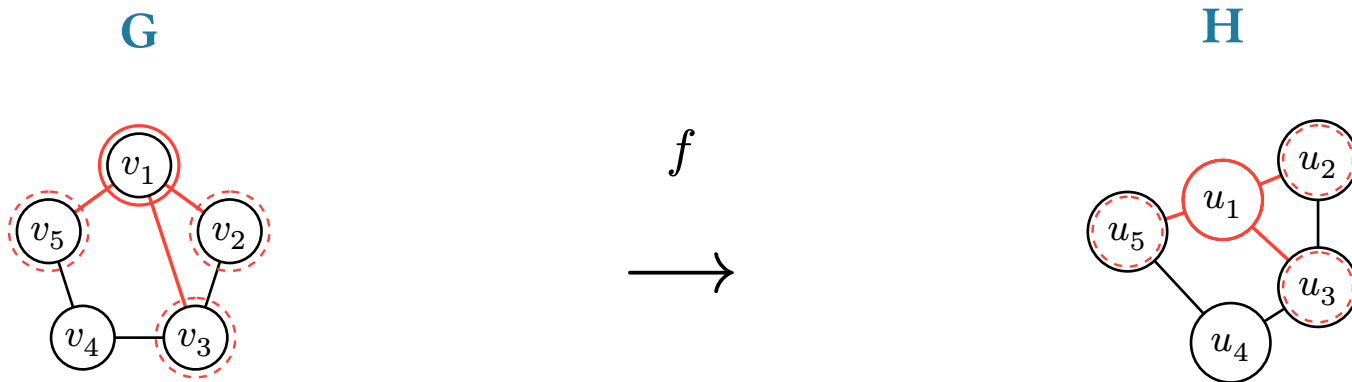
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i.e. $N_G(x) = N_G(f(x))$ for $x \in X$ where isomorphism is set of neighbours

Isomorphism ($\cong$)

Graph Isomorphism

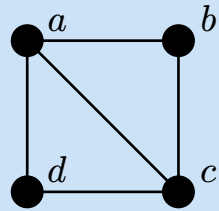
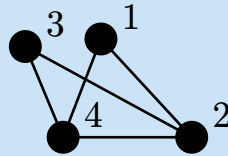
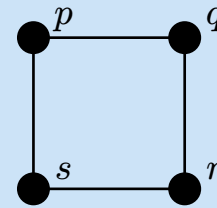
A bijection $f : V(G) \rightarrow V(H)$ with $uv \in E(G) \iff f(u)f(v) \in E(H)$ — adjacency is preserved, so neighbourhoods map onto neighbourhoods.



$$f(v_i) = u_i : \quad N(v_1) = \{v_2, v_3, v_5\} \quad \mapsto \quad N(u_1) = \{u_2, u_3, u_5\}$$

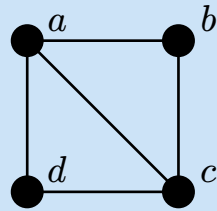
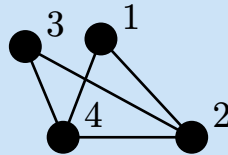
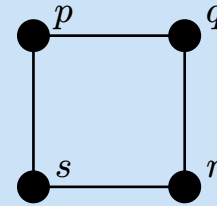
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Are the following graphs, X, Y, Z isomorphisms of each other?

 X  Y  Z

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Solution. $X \cong Y$ (both: 4 vertices, 5 edges, degrees 3, 3, 2, 2) but Z has only 4 edges.

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- **Dual of Planar Graph**: a transformation where r are represented by vertices, e form between r based on the number of edges separating them.

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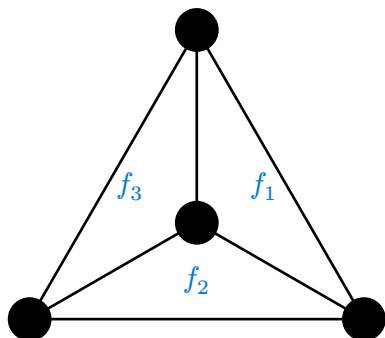
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- For a dual G^* , $|F(G)| = |V(G^*)|$ where face F is R

Dual of a Planar Graph Example

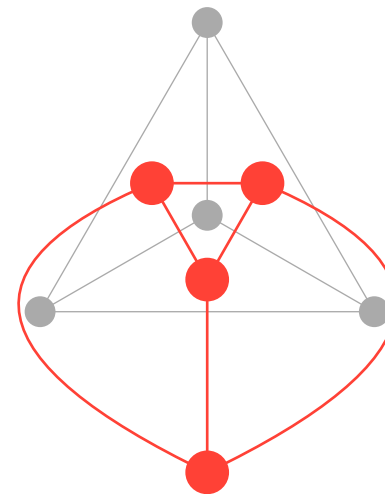
G and its 4 faces



f_4 (outer)



G^* : one vertex per face



$$|V(G^*)| = |F(G)| = 4, \quad |E(G^*)| = |E(G)| = 6$$

Planar Graphs

Euler Formula

Graph G is a connected planar multigraph with v vertices, e edges, and r regions in some planar representation if and only if,

$$v - e + r = 2.$$

Planar Graphs

Handshaking Lemma for Graphs with Regions

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Kuratowski's Theorem

G is *planar* $\iff G$ does not have $K_{3,3}$ or K_5 as a subgraph or subdivision, where a subgraph is a portion of graph G and subdivision is a graph with “extra intermediate” vertices in between the target graph ($K_{3,3}$, K_5 in this case)

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- For connected planar G with $v \geq 3$, $e \leq 3v - 6$
- For connected planar G with $v \geq 3$ and no C_3 , $e \leq 2v - 4$
- We show above two using $v - e + r = 2$ and inequality on Handshaking

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By definition the dual draws an edge between two regions based on the degree of the region so $v_{G^*} = r_G$.

Planar Graphs

Also by the handshaking lemma $2|E| = \sum_{r \in R} \deg(r)$ so the edges drawn in each dual are the same amount as in G , hence $e_{G'} = e_G$.

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Hence (using assumption), $v_{G^*} - e_{G^*} + r_{G^*} = r_G - e_G + v_G = 2$

Paths

Hamiltonian Paths

Definition

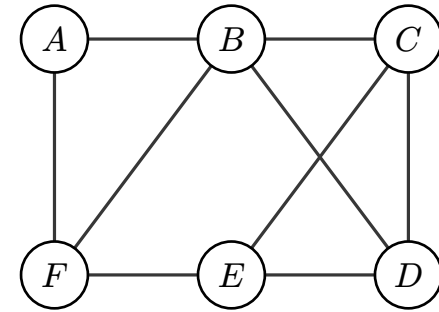
A *Hamiltonian path* is a path that visits every vertex of a graph exactly once.

Hamiltonian Paths

For a small graph, we usually use **backtracking** to find a Hamiltonian path.

1. Choose a starting vertex.
2. Move to an adjacent vertex that has not yet been visited.
3. If every vertex is visited, the path is complete.
4. If no legal move remains too early, undo the last choice and try a different edge.

A graph with six vertices



By observing, $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$ so there is a Hamilton path.

Euler Trails and Tours

Definition

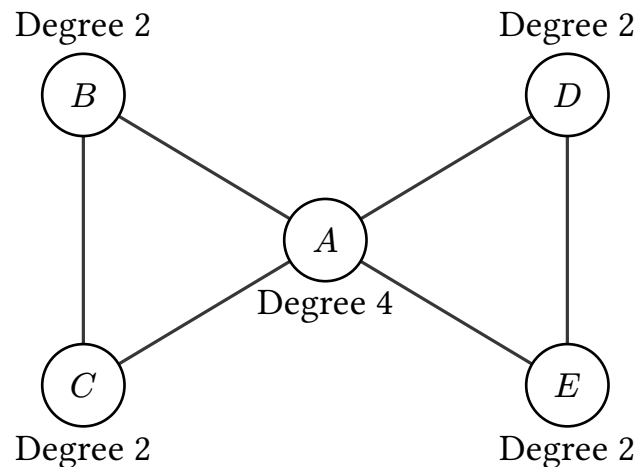
- An ***Euler trail*** travels along every edge of a graph exactly once.
- An ***Euler tour*** is an Euler trail that returns to its starting vertex.

Euler Trails and Tours

Condition of Euler Tour

- A closed **Euler tour** exists if and only if every vertex has even degree.
- An **Euler trail** with different endpoints exists if and only if there are two odd-degree vertices.

An Eulerian graph



Every vertex has even degree, so this graph has an Euler tour.

Spanning Tree

Definition

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- *If the tree minimises some weight/cost, it is a **minimum spanning tree***

Kruskal's Algorithm

Algorithm that determines a minimum spanning tree. From a graph G

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1. Select edge with min weight that doesn't link two connected vertices

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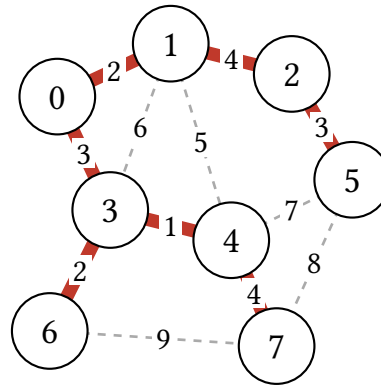
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2. Repeat 1 until all vertices connected.

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■ MST edges — total weight 19

Dijkstra's algorithm

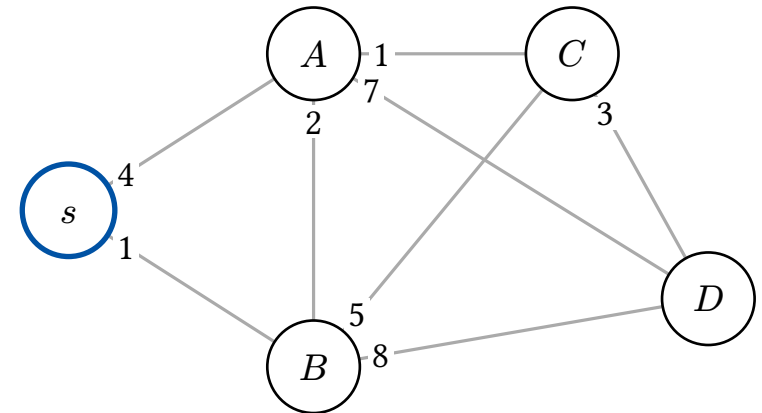
A **weighted graph** has numbers on its edges which may represent distance, time, or cost.

Define D to be the distance between two vertices s and t , Dijkstra's algorithm aims to find $\min(D(s, t))$.

Important assumption.

Every edge weight must be non-negative.

Sample graph



Dijkstra's algorithm

Dijkstra stores a **current distance**

$D_{\text{curr}}(s, u)$ the length of the best route to u found so far. Initially, $D_{\text{curr}}(s, s) = 0$ and $D_{\text{curr}}(s, u) = \infty$ ($u \neq s$). Then,

1. Choose vertex u with the smallest current distance, and mark it finished
2. Check if going through u gives a shorter route to each unfinished node

Now we update.

Suppose z is adjacent to the vertex u just chosen. Replace z 's current distance by the smaller of:

- the route already known, and
- the route that first reaches u , then uses edge $u \rightarrow z$.

Lets try it out on an example.

Dijkstra's algorithm

Initially,

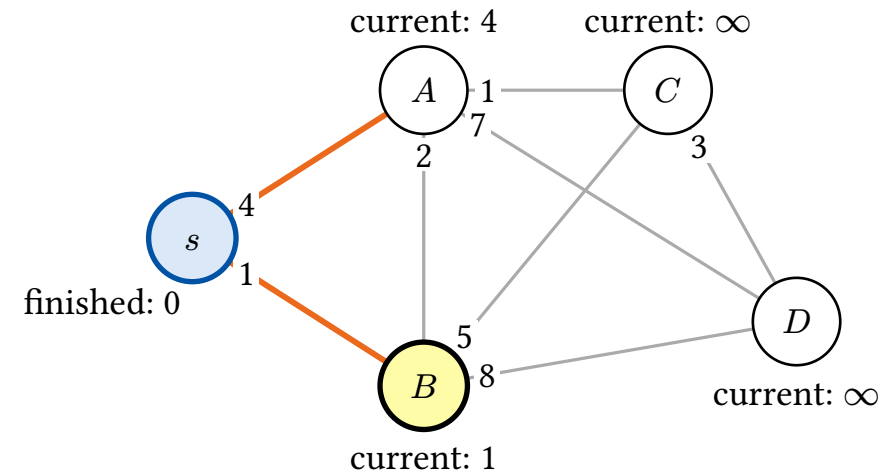
$$D_{\text{curr}}(s, s) = 0,$$

$$D_{\text{curr}}(s, A) = D_{\text{curr}}(s, B) = \infty,$$

$$D_{\text{curr}}(s, C) = D_{\text{curr}}(s, D) = \infty.$$

The smallest current distance is 0, so Dijkstra first chooses and finishes s . Checking the two edges from s gives: $D_{\text{curr}}(s, A) = 4$, $D_{\text{curr}}(s, B) = 1$. No route to C or D has been found yet, so their distances remain ∞ . The smallest unfinished value is now 1, so B will be chosen next.

After processing s



finished



chosen
next



updated
from s

Dijkstra's algorithm

Choose B , whose current distance is 1.

Routes through B give: $D_{\text{curr}}(s, A) = \min\{4, 1 + 2\} = 3$, $D_{\text{curr}}(s, C) = \min\{\infty, 1 + 5\} = 6$, $D_{\text{curr}}(s, D) = \min\{\infty, 1 + 8\} = 9$.

Choose A , whose current distance is 3.

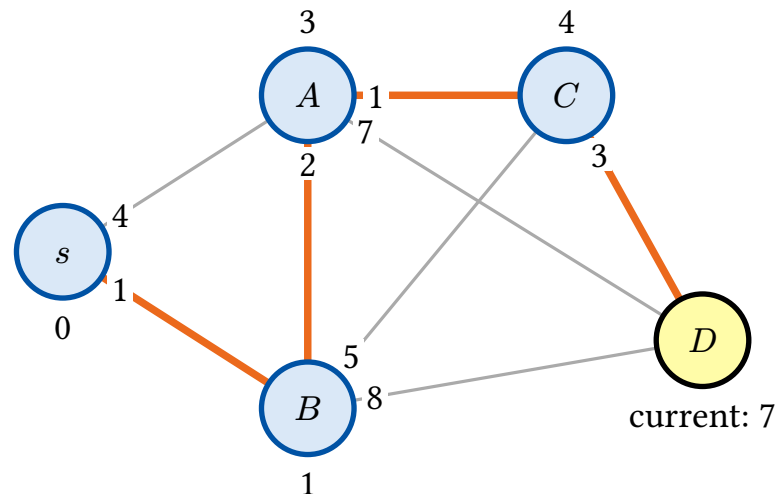
$D_{\text{curr}}(s, C) = \min\{6, 3 + 1\} = 4$, $D_{\text{curr}}(s, D) = \min\{9, 3 + 7\} = 9$.

Choose C , whose current distance is 4.

$D_{\text{curr}}(s, D) = \min\{9, 4 + 3\} = 7$.

The only unfinished vertex is now D , with current distance 7.

After processing B, A, C



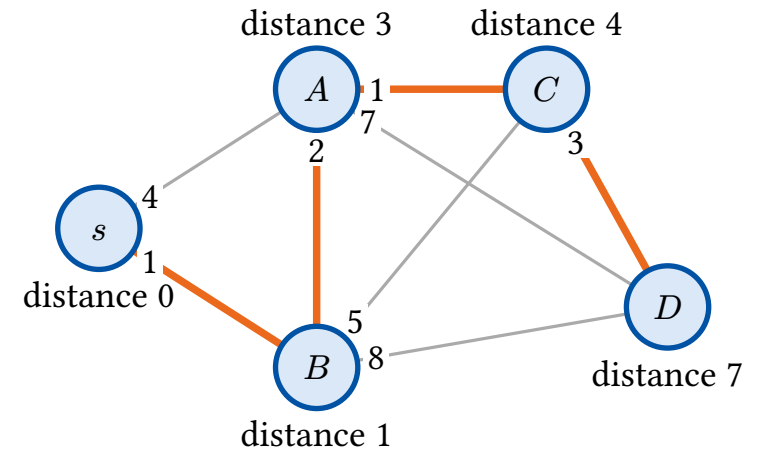
Dijkstra's algorithm

When D is chosen, every vertex is finished. The final distances and corresponding shortest paths are:

vertex	distance	one shortest path
s	0	s
B	1	$s \rightarrow B$
A	3	$s \rightarrow B \rightarrow A$
C	4	$s \rightarrow B \rightarrow A \rightarrow C$
D	7	$s \rightarrow B \rightarrow A \rightarrow C \rightarrow D$

To reconstruct a path, follow each vertex's recorded predecessor backwards until reaching s .

The shortest-path tree



Negative edge weights require a different algorithm, such as Bellman–Ford.

Cliques

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At a party of six people, every pair are either friends or strangers. Prove that there must be three people who are mutual friends or three people who are mutual strangers. (Mutual as in pairwise.)

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A clique of a graph is a set of vertices where every vertex in the clique is connected to every other vertex in the clique by edges in the graph.

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A clique of a graph G is a set of vertices $C \subseteq V(G)$ with $G[C] \cong K_{|C|}$

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Mantel's Theorem

A triangle free graph with n vertices has at most $\left\lfloor \frac{n^2}{4} \right\rfloor$ edges, with equality iff the graph is $K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}$.

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The graph induced by $V(G) - \{u, v\}$, i.e. if we take G and delete all edges connected to u or v as well as the vertices, has $n - 2$ vertices and is triangle free, so (assuming our claim holds) must have at most $\left\lfloor \frac{(n-2)^2}{4} \right\rfloor \leq \frac{(n-2)^2}{4}$ edges.

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In the extreme case, every other vertex in the graph was connected to u or v (**but not both, as otherwise we would have a triangle**). So we could have deleted at most $\deg(u) + \deg(v) - 1 \leq n - 1$ edges (uv counted twice).

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Hence, $|E(G)| \leq \frac{(n-2)^2}{4} + \deg(u) + \deg(v) - 1 \leq \frac{(n-2)^2}{4} + n - 1 = \frac{n^2}{4}$.

Mantel's Theorem

Problem. (**AOPS**)

In a chess tournament there are 670 participants, and no two participants compete against each other more than once. After a series of matches, it is known that whenever any three participants are chosen, there is at least one pair of two players among them that have competed against one another.

Find the minimal amount of matches that would have had to take place in order for this to be possible.

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Solution. We will prove this for $2n$ participants. Construct a graph G where every participant is a vertex, and an edge is present between any two participants iff they have played a match against each other.

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Our condition turns into $\overline{G}$ being triangle free. By Mantel's theorem, we have $|E(\overline{G})| \leq \frac{(2n)^2}{4}$, so $|E(G)| \geq |K_{2n}| - n^2 = n(n-1)$.

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To show this is minimal, we need a construction. Notice this is $2\binom{n}{2}$, so obvious way is to split the participants into two groups and have a match between every pair in a group, which works. (Note that $\overline{G} \cong K_{n,n}$, equality case)

Turan's Theorem

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A graph with n vertices whose cliques have at most r vertices has at most

$$\left\lfloor \left(1 - \frac{1}{r}\right) \frac{n^2}{2} \right\rfloor$$

edges. Mantel's theorem is the case where $r = 2$.

Proof: By induction, consider the graph induced by $V(G) - \{v_1, v_2, \dots, v_r\}$. Same argument as above, details are left to the reader 😊.

Turan's Theorem

The bound is only always achievable for $r \leq 7$. To be exact, there is a family of graphs called the **Turan Graphs** which maximise the number of edges, and it can be shown that these graphs have

$$\left(1 - \frac{1}{r} - o(1)\right) \frac{n^2}{2}$$

edges, so the bound is asymptotically correct. Details can be found on the Wikipedia page for Turan Graphs or MATH5505.

Matchings

Matching Fundamentals

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- If every $v \in G$ has an edge in M , then a matching M is **perfect**

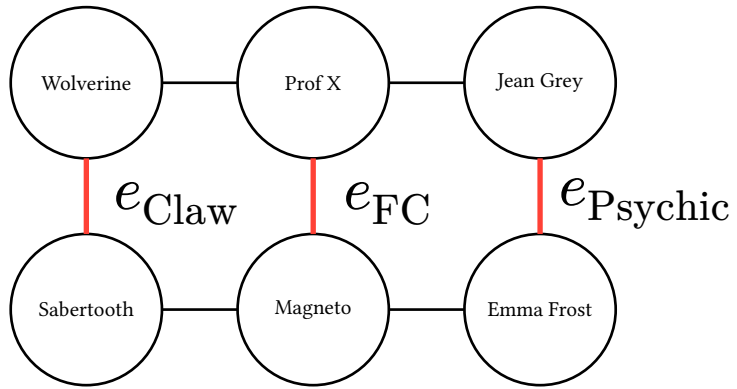
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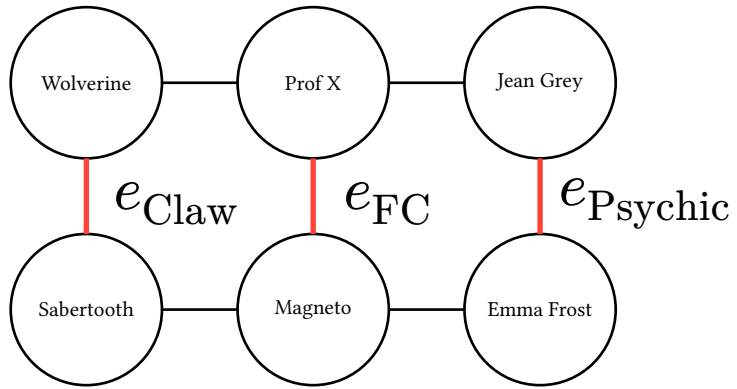
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- If every $v \in G$ has an edge in M , then a matching M is **perfect**
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- M is **perfect** $\Leftrightarrow |\#M\text{-exposed vertices}| = 0$

Matching Fundamentals



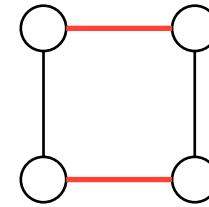
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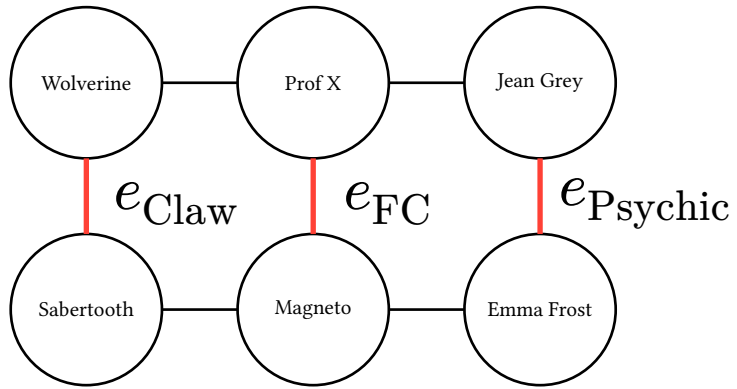


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Perfect Matching (all v)

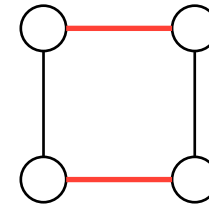


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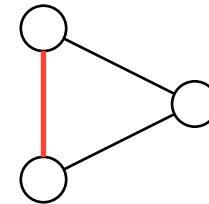


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Perfect Matching (all v)



M-exposed Matching



Hall's Theorem

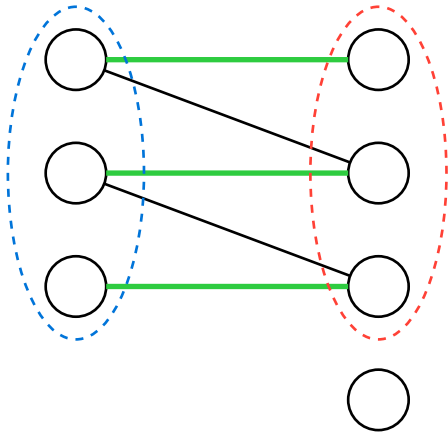
Hall's (Marriage) Theorem

Let $G = (X, Y, E)$ be a bipartite graph where the edges $E = e(G)$ only connect $x \in X$ to $y \in Y$.

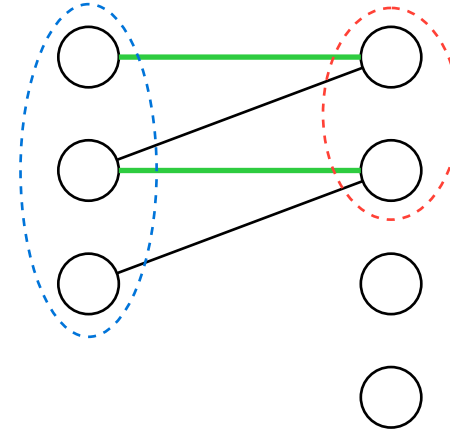
*We define for $S \subseteq X$, $N_G(S)$ as the set of vertices in Y **adjacent** to X .*

*Then there exists a **perfect** matching M for all $S \subseteq X \Leftrightarrow |\Gamma(S)| \geq |S|$.*

Hall's Theorem



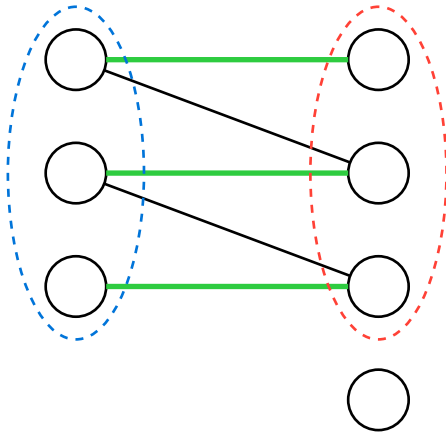
$$|N_G(S)| \geq |S|$$



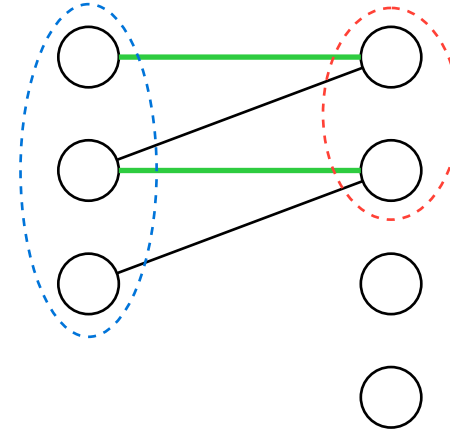
$$|N_G(S)| < |S|$$

(Hall's condition fails)

Hall's Theorem



$$|N_G(S)| \geq |S|$$



$$|N_G(S)| < |S|$$

(Hall's condition fails)

Every “apprentice” who wants “instructor” should have one vice-versa.

Application of Hall's Theorem

Problem. (*Canadian MO 2006*) In a rectangular array of nonnegative real numbers with m rows and n columns, each row and each column contains at least one positive element. Moreover, if a row and a column intersect at a positive element, then the sums of their elements are the same. Prove $m = n$.

Appendix

That's All Folks!

Thanks for coming! We hope you enjoyed 😊, pizza is coming.

Please complete the
feedback form 🙄

More mathematics
workshops to come!!!

Feel free to try the
worksheets and discuss!!!



Feedback Form

